

Introduction

Reactome is open-source, open access, manually curated and peer-reviewed pathway database. Pathway annotations are authored by expert biologists, in collaboration with Reactome editorial staff and cross-referenced to many bioinformatics databases. A system of evidence tracking ensures that all assertions are backed up by the primary literature. Reactome is used by clinicians, geneticists, genomics researchers, and molecular biologists to interpret the results of high-throughput experimental studies, by bioinformaticians seeking to develop novel algorithms for mining knowledge from genomic studies, and by systems biologists building predictive models of normal and disease variant pathways.

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Literature references

Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A. et al. (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC bioinformatics*, 18, 142. [↗](#)

Sidiropoulos, K., Sidiropoulos, K., Sidiropoulos, K., Sidiropoulos, K., Sidiropoulos, K., Sidiropoulos, K. et al. (2017). Reactome enhanced pathway visualization. *Bioinformatics*, 33, 3461-3467. [↗](#)

Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A. et al. (2018). The Reactome Pathway Knowledgebase. *Nucleic Acids Res*, 46, D649-D655. [↗](#)

Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A., Fabregat, A. et al. (2018). Reactome graph database: Efficient access to complex pathway data. *PLoS computational biology*, 14, e1005968. [↗](#)

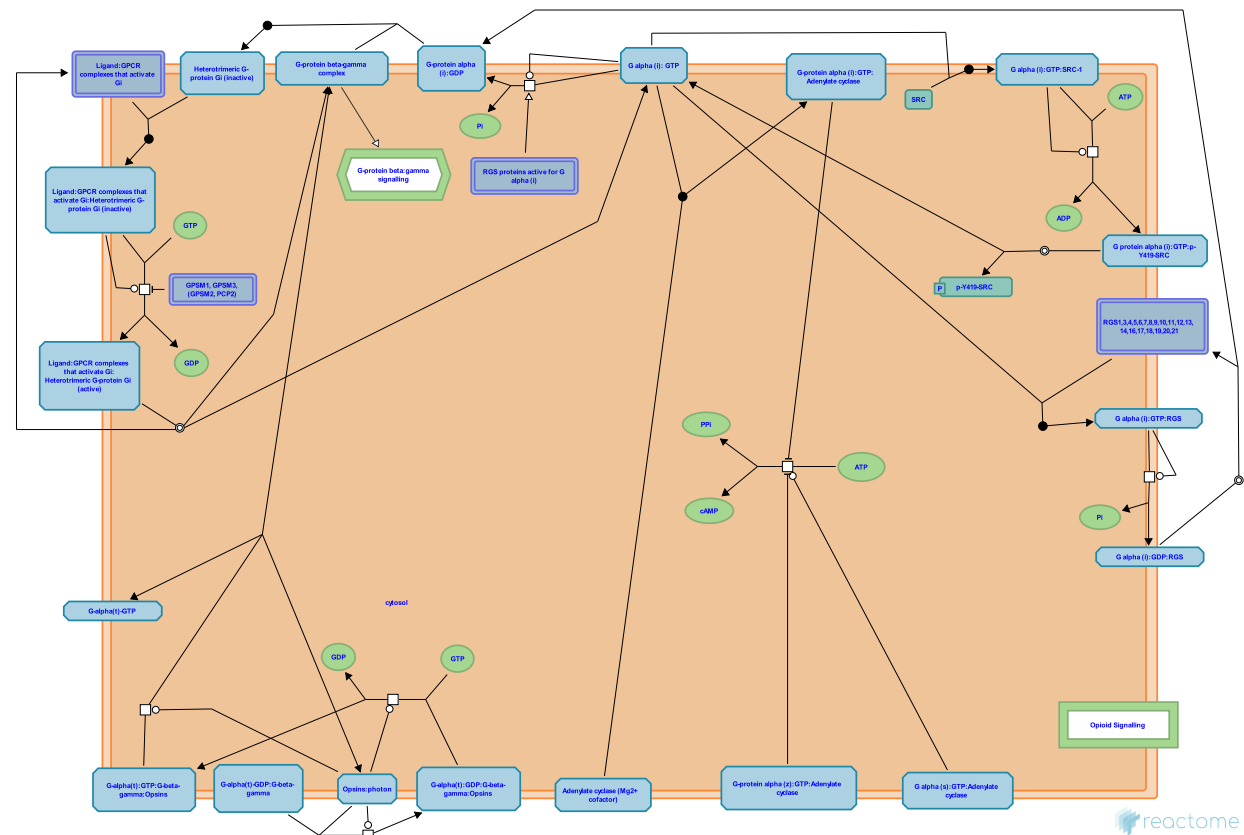
Reactome database release: 95

This document contains 2 pathways and 16 reactions ([see Table of Contents](#))

G alpha (i) signalling events ↗

Stable identifier: R-HSA-418594

Compartments: plasma membrane



The classical signalling mechanism for G alpha (i) is inhibition of the cAMP dependent pathway through inhibition of adenylyl cyclase (Dessauer C W et al. 2002). Decreased production of cAMP from ATP results in decreased activity of cAMP-dependent protein kinases. Other functions of G alpha (i) includes activation of the protein tyrosine kinase c-Src (Ma Y C et al. 2000). Regulator of G-protein Signalling (RGS) proteins can regulate the activity of G alpha (i) (Soundararajan et al. 2008).

Literature references

Hildebrandt, JD. (1997). Role of subunit diversity in signaling by heterotrimeric G proteins. *Biochem Pharmacol*, 54, 325-39. ↗

Gilman, AG. (1987). G proteins: transducers of receptor-generated signals. *Annu Rev Biochem*, 56, 615-49. ↗

Editions

2009-05-01	Authored, Edited	Jupe, S.
2009-06-03	Reviewed	Akkerman, JW.
2017-07-10	Revised	Varusai, TM.

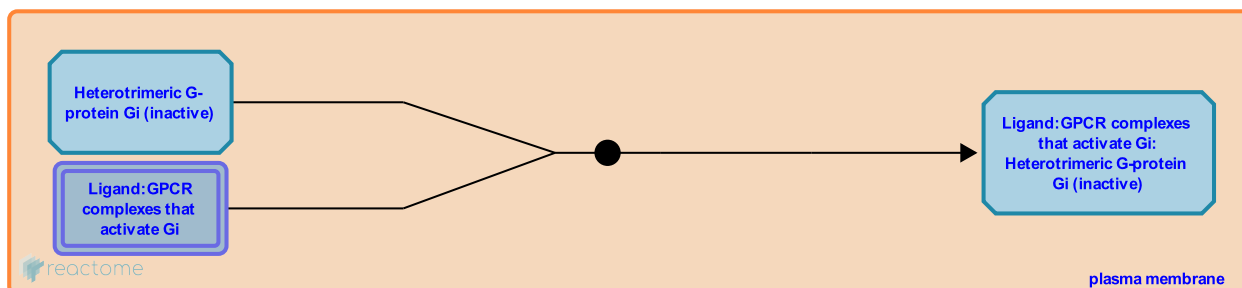
Liganded Gi-activating GPCRs bind inactive heterotrimeric G-protein Gi ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-749456

Type: binding

Compartments: plasma membrane



Many unrelated GPCRs couple with the Gi G-protein subtype. The G-alpha (i) subunit inhibits the production of cAMP from ATP. In turn, this results in decreased activity of cAMP-dependent protein kinase. There are 8 types of G-alpha (i) known to date: G(i)1, G(i)2, G(i)3, G(i)o, G(i)z, G(i)gust (gustducin) and two G(i)t (retinal transducin) (Downes GB and Gautam N, 1999). Once GDP is exchanged for GTP on the alpha subunit, it dissociates from the G-beta-gamma subunit.

Followed by: [Liganded Gi-activating GPCR acts as a GEF for Gi](#)

Literature references

Itoh, H., Toyama, R., Kozasa, T., Tsukamoto, T., Matsuoka, M., Kaziro, Y. (1988). Presence of three distinct molecular species of Gi protein alpha subunit. Structure of rat cDNAs and human genomic DNAs. *J Biol Chem*, 263, 6656-64. ↗

Editions

2010-05-18	Authored	Jupe, S.
2010-05-22	Reviewed	D'Eustachio, P.
2010-05-26	Edited	Jupe, S.
2017-07-10	Revised	Varusai, TM.

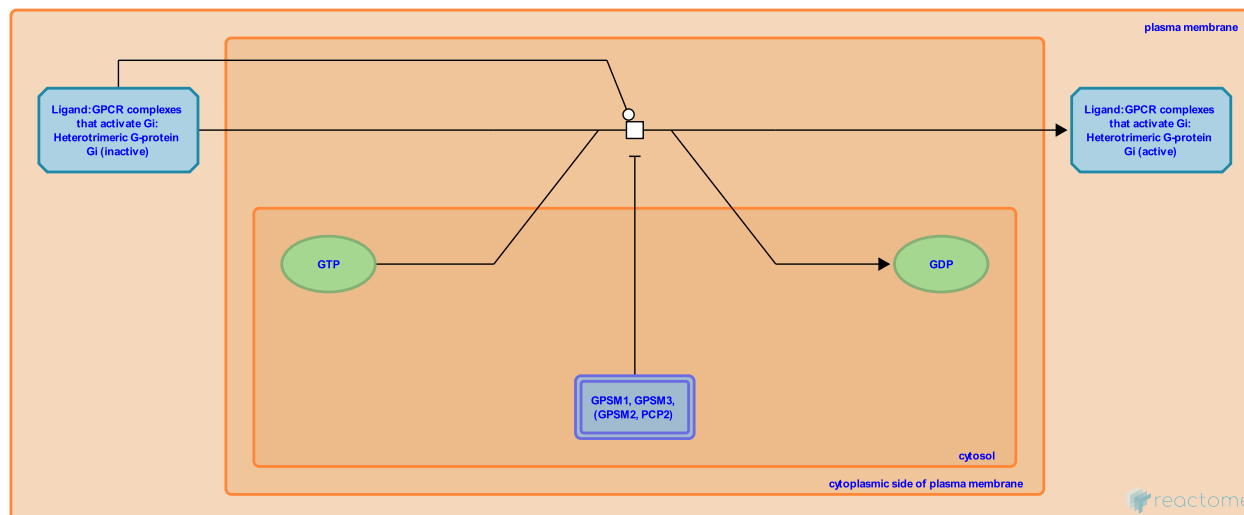
Liganded Gi-activating GPCR acts as a GEF for Gi ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-380073

Type: transition

Compartments: cytoplasmic side of plasma membrane



The liganded receptor undergoes a conformational change, generating a signal that is propagated in a manner that is not completely understood to the G-protein. This stimulates the exchange of GDP for GTP in the G-protein alpha subunit, activating the G-protein.

This event is negatively regulated by some Activators of G protein signaling (AGS) proteins, a class of proteins identified in yeast functional screens for proteins able to activate G protein signaling in the absence of a G protein-coupled receptor (GPCR) (Cismowski et al. 1999, Takesono et al. 1999). AGS proteins contain G protein regulatory (GPR) motifs (also referred to as the GoLoco motif) that bind and stabilize the Galpha subunit in its GDP-bound conformation (Mochizuki et al. 1996, Peterson et al. 2000, Cao et al. 2004, Blumer & Lanier 2014). Some RGS proteins similarly bind to Galpha preventing the exchange of GDP for GTP (Soundararajan et al. 2008).

Preceded by: [Liganded Gi-activating GPCRs bind inactive heterotrimeric G-protein Gi](#)

Followed by: [The Ligand:GPCR:Gi complex dissociates](#)

Literature references

- Itoh, H., Toyama, R., Kozasa, T., Tsukamoto, T., Matsuoka, M., Kaziro, Y. (1988). Presence of three distinct molecular species of Gi protein alpha subunit. Structure of rat cDNAs and human genomic DNAs. *J Biol Chem*, 263, 6656-64. ↗
- Takami, S., Getchell, TV., McLaughlin, SK., Margolskee, RF., Getchell, ML. (1994). Human taste cells express the G protein alpha-gustducin and neuron-specific enolase. *Brain Res Mol Brain Res*, 22, 193-203. ↗
- Lerea, CL., Bunt-Milam, AH., Hurley, JB. (1989). Alpha transducin is present in blue-, green-, and red-sensitive cone photoreceptors in the human retina. *Neuron*, 3, 367-76. ↗
- Van Dop, C., Medynski, DC., Apone, LM. (1989). Nucleotide sequence for a cDNA encoding the alpha subunit of retinal transducin (GNAT1) isolated from the human eye. *Nucleic Acids Res*, 17, 4887. ↗

Editions

2008-11-07	Authored	Jassal, B.
2008-11-29	Reviewed	D'Eustachio, P.
2017-07-10	Revised	Varusai, TM.

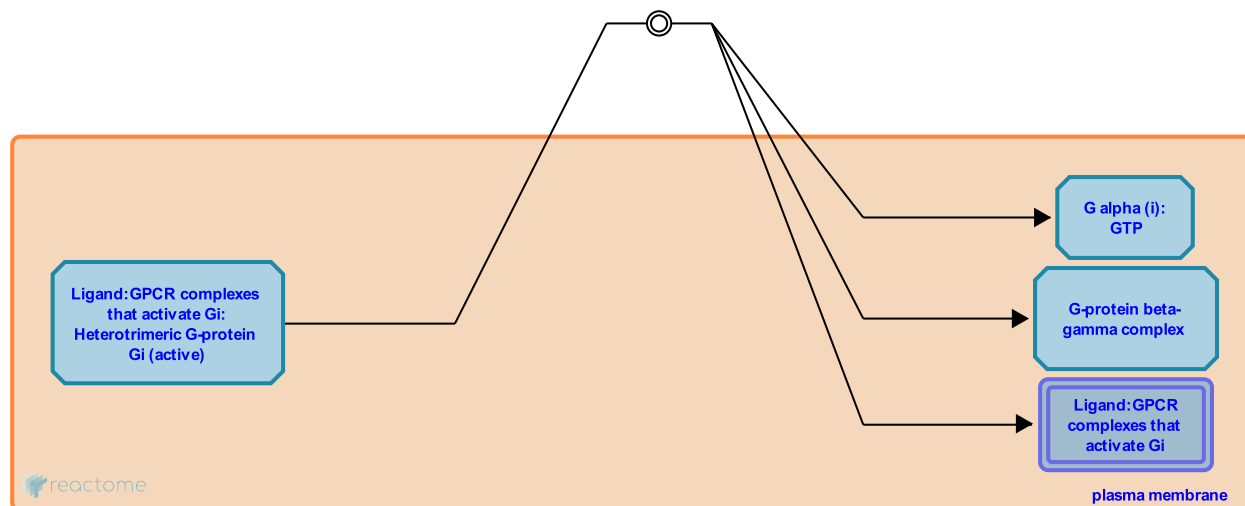
The Ligand:GPCR:Gi complex dissociates ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-749454

Type: dissociation

Compartments: extracellular region, plasma membrane



The classical view of G-protein signalling is that the G-protein alpha subunit dissociates from the beta:gamma dimer. Activated G alpha (i) and the beta:gamma dimer then participate in separate signaling cascades. Although G protein dissociation has been contested (e.g. Bassi et al. 1996), recent in vivo experiments have demonstrated that dissociation does occur, though possibly not to completion (Lambert 2008).

Preceded by: [Liganded Gi-activating GPCR acts as a GEF for Gi](#)

Followed by: [Active G alpha \(i\)-i1/i2/i3 binds RGS proteins](#), [G alpha \(i\):GTP binds SRC](#), [G alpha \(i\) auto-inactivates by hydrolysing GTP to GDP](#), [G alpha \(i\) inhibits adenylate cyclase](#)

Literature references

Lambert, NA. (2008). Dissociation of heterotrimeric g proteins in cells. *Sci Signal*, 1, re5. ↗

Editions

2010-05-18	Authored	Jupe, S.
2010-05-22	Reviewed	D'Eustachio, P.
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G alpha (i) inhibits adenylate cyclase ↗

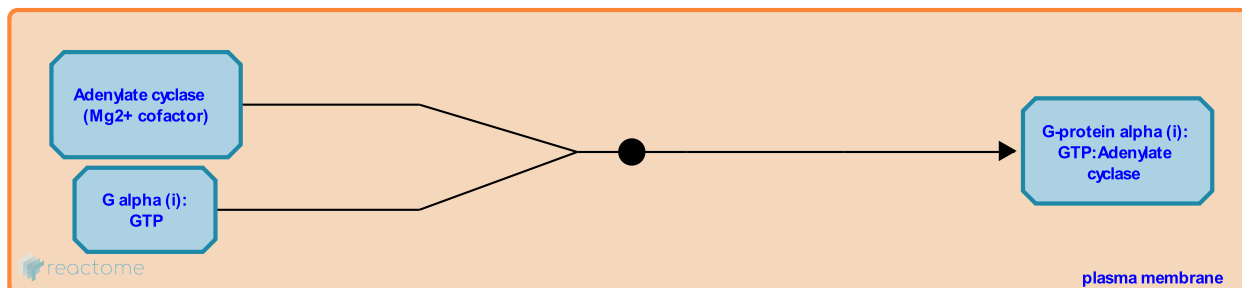
Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-392206

Type: binding

Compartments: plasma membrane

Inferred from: [G alpha \(i\) inhibits adenylate cyclase \(Canis familiaris\)](#)



G-proteins in the Gi class inhibit adenylate cyclase activity, decreasing the production of cAMP from ATP, which has many consequences but classically results in decreased activity of Protein Kinase A (PKA). cAMP also activates the cyclic nucleotide-gated ion channels, a process that is particularly important in olfactory cells.

Preceded by: [The Ligand:GPCR:Gi complex dissociates](#)

Literature references

Taussig, R., Iñiguez-Lluhi, JA., Gilman, AG. (1993). Inhibition of adenylyl cyclase by Gi alpha. *Science*, 261, 218-21. ↗

Taussig, R., Tang, WJ., Hepler, JR., Gilman, AG. (1994). Distinct patterns of bidirectional regulation of mammalian adenylyl cyclases. *J Biol Chem*, 269, 6093-100. ↗

Dessauer, CW., Chen-Goodspeed, M., Chen, J. (2002). Mechanism of Galpha i-mediated inhibition of type V adenylyl cyclase. *J. Biol. Chem.*, 277, 28823-9. ↗

Editions

2009-02-26	Authored	Jupe, S.
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Adenylate cyclase converts ATP to 3',5'-cyclic AMP (cAMP) and pyrophosphate ↗

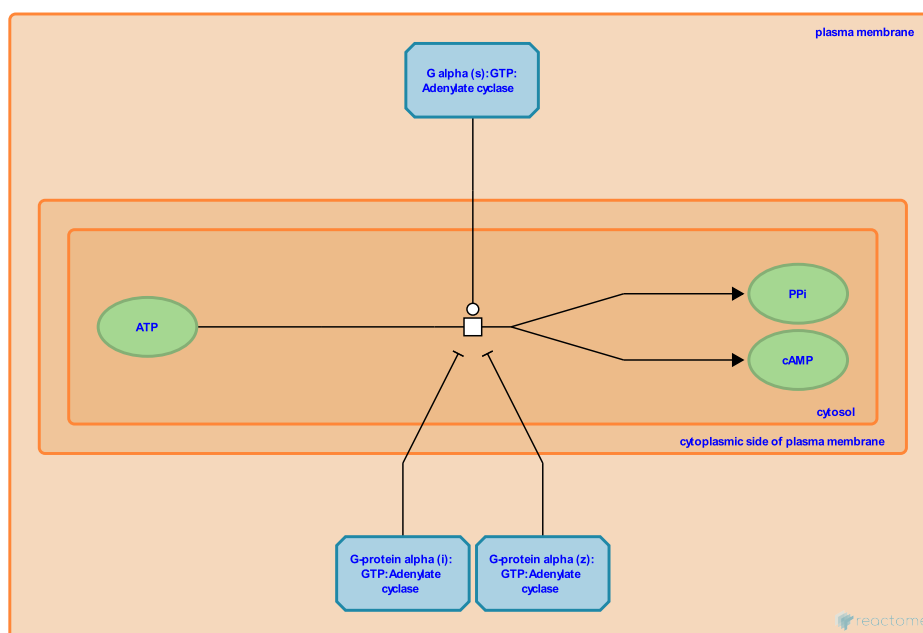
Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-392129

Type: transition

Compartments: cytosol, plasma membrane

Inferred from: [Adenylate cyclase converts ATP to 3',5'-cyclic AMP \(cAMP\) and pyrophosphate \(Canis familiaris\)](#)



The activation of adenylyl (adenylate) cyclase (AC) results in the production of adenosine-3',5'-monophosphate i.e. cyclic AMP. Humans have 9 genes encoding membrane-associated AC and one encoding a soluble AC. Two of the classes of heterotrimeric G-proteins are named according to their effect on AC; G(s) stimulates all membrane-bound ACs (the s in G(s) denotes AC stimulatory); the G(i) class inhibits some AC isoforms, particularly 5 and 6. Beta-gamma subunits of heterotrimeric G-proteins can also regulate AC. Ca²⁺/Calmodulin activates some AC isoforms (1, 8 and 3) but is inhibitory to others (5 and 6).

Literature references

SUTHERLAND, EW., RALL, TW. (1958). Fractionation and characterization of a cyclic adenine ribonucleotide formed by tissue particles. *J Biol Chem*, 232, 1077-91. ↗

Editions

2009-05-22	Authored	Jupe, S.
2009-06-03	Reviewed	Akkerman, JW.
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2017-07-10	Revised	Varusai, TM.

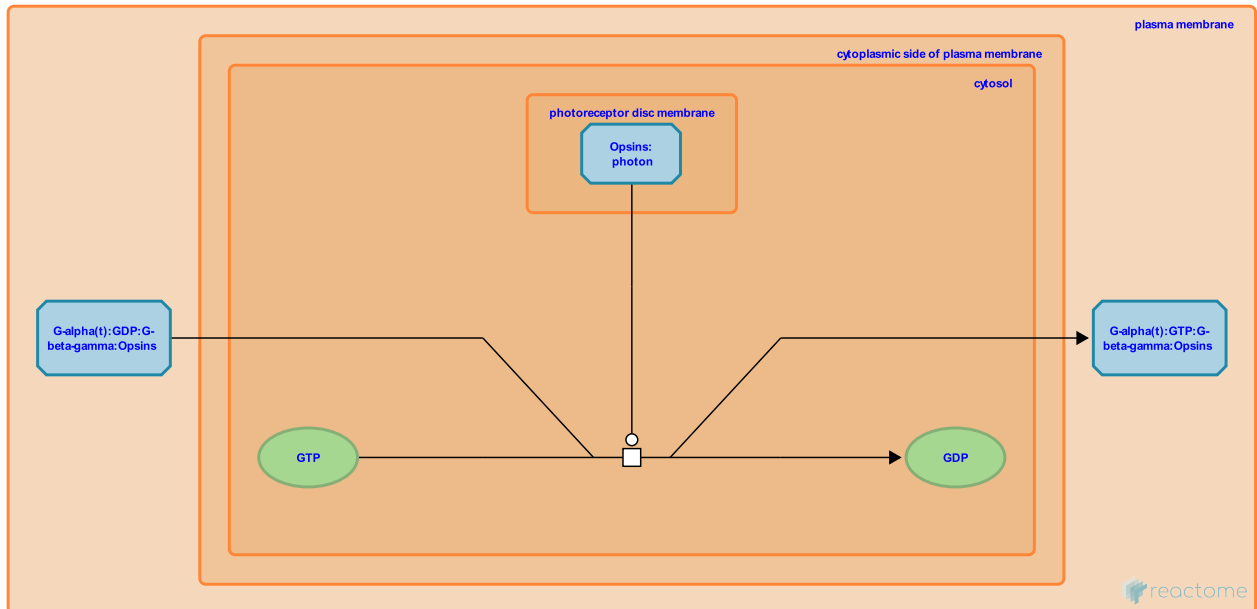
Opsins act as GEFs for G alpha-t ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-420883

Type: transition

Compartments: cytosol, plasma membrane



Transducin (Gt) is a heterotrimeric G protein encoded by GNAT genes and is naturally expressed in vertebrate retina rods and cones. There are two types, alpha-1 chain (expressed in rods by GNAT1) (Lerea CL et al, 1989) and alpha-2 chain (expressed in cones by GNAT2) (Morris TA and Fong SL, 1993). Stimulated opsins can act as GEFs for G (t) alpha subunits by replacing GDP with GTP. Consequently, the G (t) alpha subunit is activated and results in the "vertebrate phototransduction cascade" (Chen CK, 2005). Cyclic GMP Phosphodiesterase is activated which lowers cGMP levels (an intracellular second messenger molecule). Lower cGMP levels can then lead to the closure of cGMP-regulated Na⁺ and Ca²⁺ ion channels and a hyperpolarized membrane potential.

Preceded by: [Opsins binds G alpha-t](#)

Followed by: [G-alpha\(t\):G-beta-gamma:Opsins complex dissociates to active G-alpha\(t\)](#)

Literature references

- Lerea, CL., Bunt-Milam, AH., Hurley, JB. (1989). Alpha transducin is present in blue-, green-, and red-sensitive cone photoreceptors in the human retina. *Neuron*, 3, 367-76. ↗
- Morris, TA., Fong, SL. (1993). Characterization of the gene encoding human cone transducin alpha-subunit (GNAT2). *Genomics*, 17, 442-8. ↗
- Chen, CK. (2005). The vertebrate phototransduction cascade: amplification and termination mechanisms. *Rev Physiol Biochem Pharmacol*, 154, 101-21. ↗

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2009-05-14	Authored	Jassal, B.
2009-05-29	Reviewed	D'Eustachio, P.
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G alpha (i) auto-inactivates by hydrolysing GTP to GDP ↗

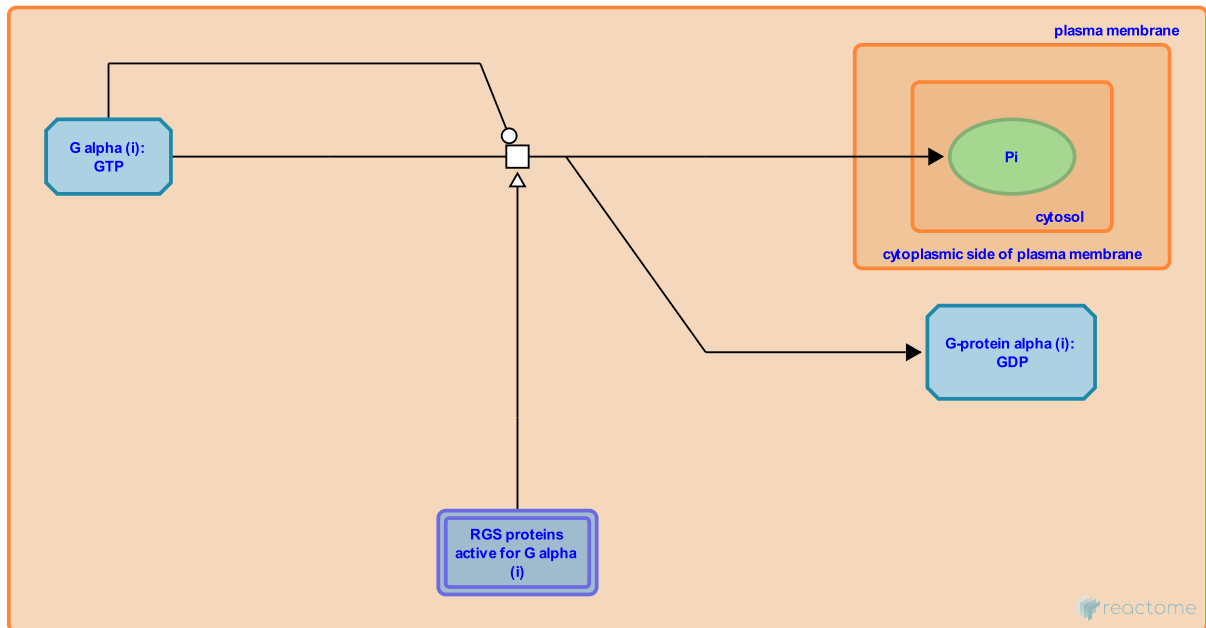
Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-392212

Type: transition

Compartments: plasma membrane

Inferred from: [G alpha \(i\)1 auto-inactivates by hydrolysing GTP to GDP \(Rattus norvegicus\)](#)



When a ligand activates a G protein-coupled receptor, it induces a conformational change in the receptor (a change in shape) that allows the receptor to function as a guanine nucleotide exchange factor (GEF), stimulating the exchange of GDP for GTP on the G alpha subunit. In the traditional view of heterotrimeric protein activation, this exchange triggers the dissociation of the now active G alpha subunit from the beta:gamma dimer, initiating downstream signalling events. The G alpha subunit has intrinsic GTPase activity and will eventually hydrolyze the attached GTP to GDP, allowing reassociation with G beta:gamma. Additional GTPase-activating proteins (GAPs) stimulate the GTPase activity of G alpha, leading to more rapid termination of the transduced signal. In some cases the downstream effector may have GAP activity, helping to deactivate the pathway. This is the case for phospholipase C beta, which possesses GAP activity within its C-terminal region (Kleuss et al. 1994).

Preceded by: [The Ligand:GPCR:Gi complex dissociates](#)

Followed by: [Inactive G alpha \(i\) reassociates with G beta:gamma](#)

Literature references

Kleuss, C., Raw, AS., Lee, E., Sprang, SR., Gilman, AG. (1994). Mechanism of GTP hydrolysis by G-protein alpha subunits. *Proc Natl Acad Sci U S A*, 91, 9828-31. ↗

Editions

2009-02-27	Authored	Jupe, S.
2009-06-03	Reviewed	Akkerman, JW.
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2017-07-10	Revised	Varusai, TM.

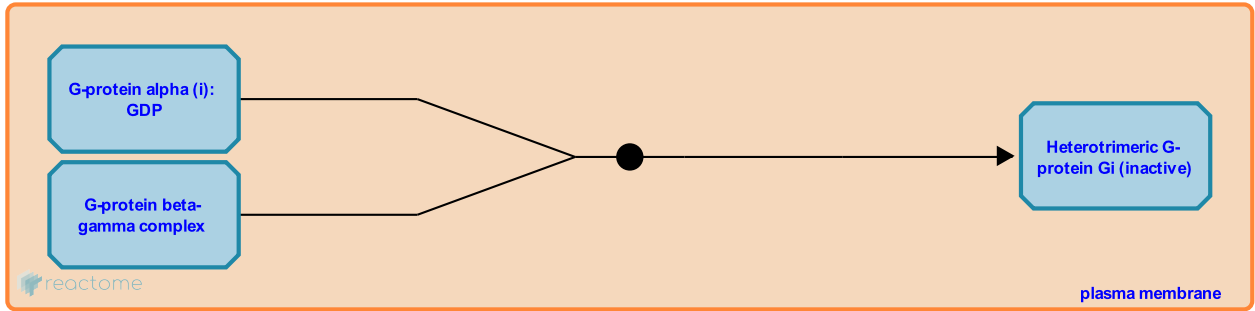
Inactive G alpha (i) reassociates with G beta:gamma ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-751001

Type: binding

Compartments: plasma membrane



The classical model of G-protein signaling suggests that the G-protein dissociates upon GPCR activation. The active G alpha (i) subunit then participates in signaling, until its intrinsic GTPase activity degrades the bound GTP to GDP. The inactive G alpha (i):GDP complex has much higher affinity for the G beta:gamma complex and consequently reassociates.

Preceded by: [G alpha \(i\) auto-inactivates by hydrolysing GTP to GDP](#)

Literature references

Oldham, WM., Hamm, HE. (2006). Structural basis of function in heterotrimeric G proteins. *Q Rev Biophys*, 39, 117-66 . ↗

Siderovski, DP., Willard, FS. (2005). The GAPs, GEFs, and GDIs of heterotrimeric G-protein alpha subunits. *Int J Biol Sci*, 1, 51-66. ↗

Editions

2010-05-18	Authored	Jupe, S.
2010-05-22	Reviewed	D'Eustachio, P.
2010-05-26	Edited	Jupe, S.
2017-07-10	Revised	Varusai, TM.

G alpha (i):GTP binds SRC ↗

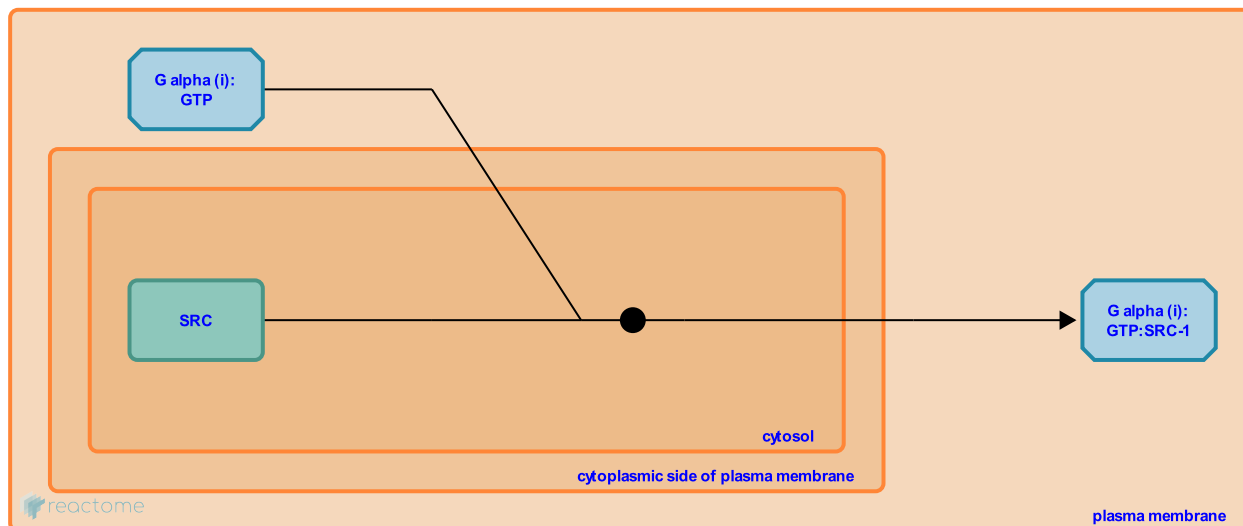
Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8964311

Type: binding

Compartments: cytosol, plasma membrane

Inferred from: [G alpha \(i\):GTP binds Src \(Mus musculus\)](#)



G-Protein Coupled Receptors (GPCR) sense extracellular signals and activate different Guanine nucleotide binding proteins. Upon activation, the Guanine nucleotide-binding protein G(i) subunit alpha (GNAI) can bind directly to proto-oncogene tyrosine-protein kinase Src (SRC) in vitro and in vivo. This binding results in autophosphorylation of SRC and subsequently its activation. Physiologically, the SRC family have implications in cell growth and cancer.

Preceded by: [The Ligand:GPCR:Gi complex dissociates](#)

Followed by: [G protein alpha \(i\)-SRC complex catalyzes SRC to p-Y419-SRC](#)

Literature references

Ma, YC., Huang, J., Ali, S., Lowry, W., Huang, XY. (2000). Src tyrosine kinase is a novel direct effector of G proteins. *Cell*, 102, 635-46. ↗

Editions

2017-07-27	Authored, Edited	Varusai, TM.
2018-09-13	Reviewed	Huang, X.

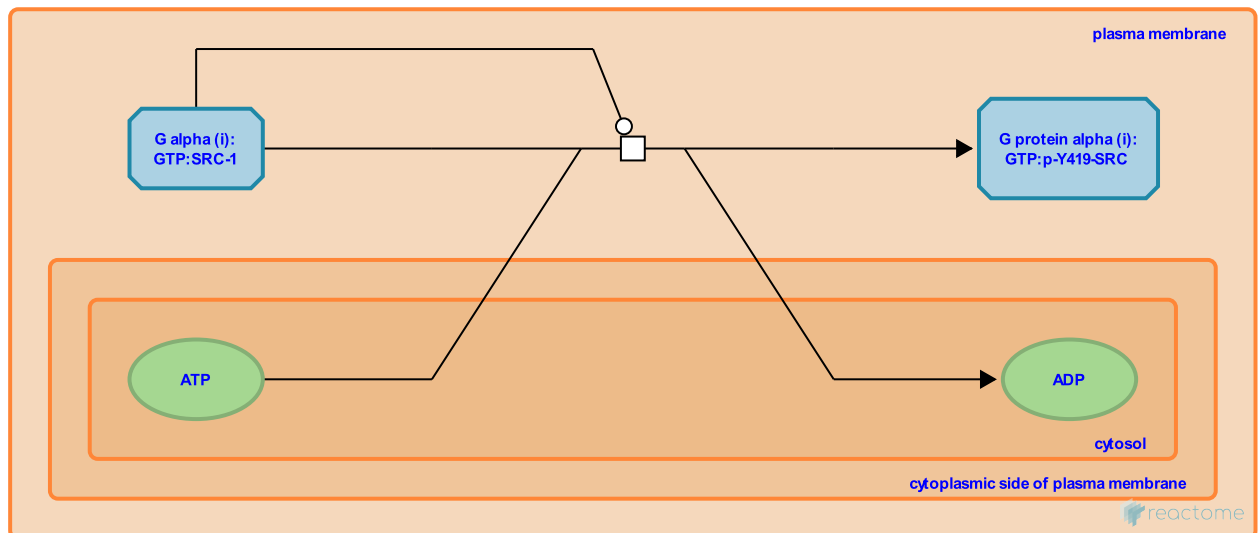
G protein alpha (i)-SRC complex catalyzes SRC to p-Y419-SRC ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8964242

Type: transition

Compartments: plasma membrane



G-Protein Coupled Receptors sense extracellular signals and activate different Guanine nucleotide binding proteins. Upon activation, the alpha subunit of the G protein (GNAI) can directly bind to SRC. In the presence of active GNAI, SRC can autophosphorylate the 419-tyrosine residue, which leads to the subsequent activation of SRC. Physiologically, the SRC family have implications in cell growth and cancer.

Preceded by: [G alpha \(i\):GTP binds SRC](#)

Followed by: [G protein alpha \(i\)-SRC complex dissociates to G protein alpha \(i\) and Active SRC](#)

Literature references

Ma, YC., Huang, J., Ali, S., Lowry, W., Huang, XY. (2000). Src tyrosine kinase is a novel direct effector of G proteins. *Cell*, 102, 635-46. ↗

Editions

2017-07-27	Authored, Edited	Varusai, TM.
2018-09-13	Reviewed	Huang, X.

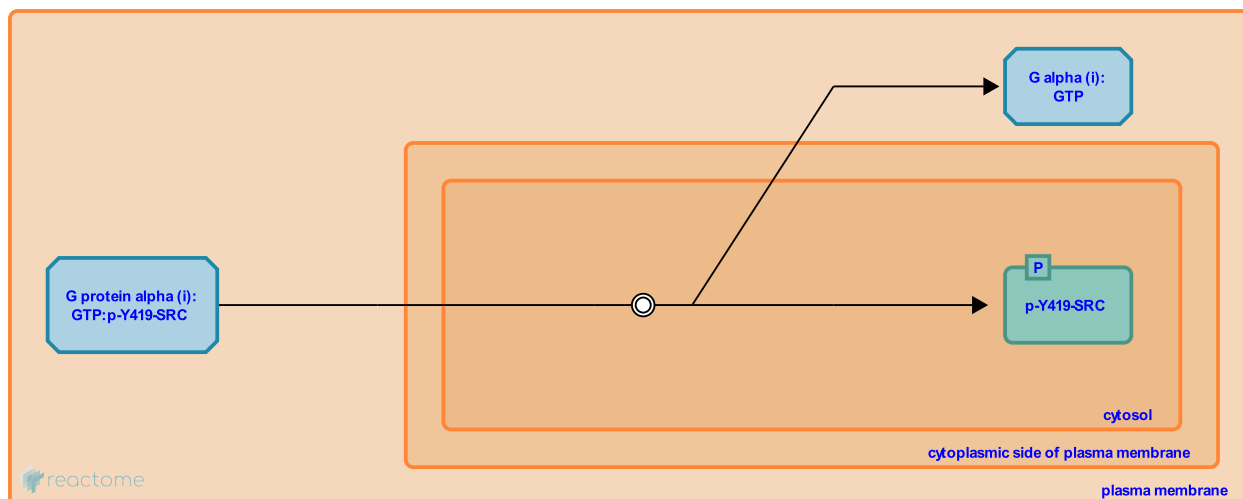
G protein alpha (i)-SRC complex dissociates to G protein alpha (i) and Active SRC ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8964329

Type: dissociation

Compartments: cytosol, plasma membrane



G-Protein Coupled Receptors sense extracellular signals and activate different Guanine nucleotide binding proteins. Upon activation, the Guanine nucleotide-binding protein G(i) subunit alpha (GNAI) can directly bind to proto-oncogene tyrosine-protein kinase Src (SRC). When bound to active GNAI, SRC can autophosphorylate the 416-tyrosine residue. Upon autophosphorylation, SRC dissociates from GNAI and is subsequently activated. GNAI increases SRC enzymatic activity by decreasing K_m and not affecting V_m . Physiologically, the SRC family have implications in cell growth and cancer.

Preceded by: [G protein alpha \(i\)-SRC complex catalyzes SRC to p-Y419-SRC](#)

Literature references

Ma, YC., Huang, J., Ali, S., Lowry, W., Huang, XY. (2000). Src tyrosine kinase is a novel direct effector of G proteins. *Cell*, 102, 635-46. ↗

Editions

2017-07-27	Authored, Edited	Varusai, TM.
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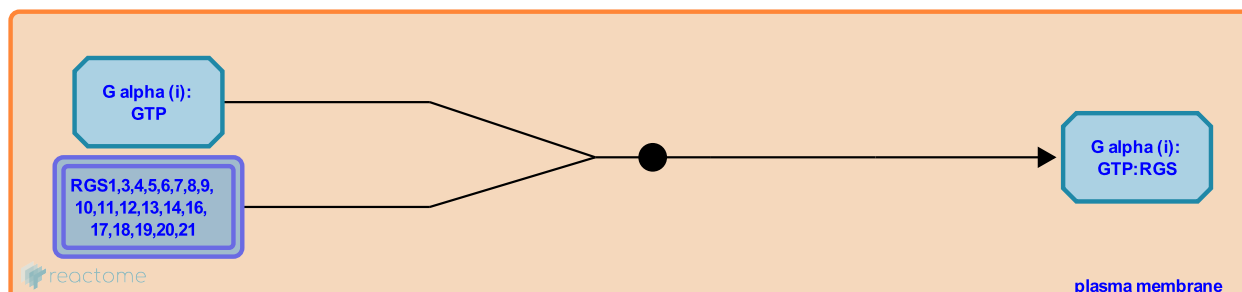
Active G alpha (i)-i1/i2/i3 binds RGS proteins ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8982019

Type: binding

Compartments: plasma membrane, cytosol



G Protein Coupled Receptors (GPCR) sense extracellular signals and activate different Guanine nucleotide binding proteins (G proteins). Upon activation, GPCRs can replace the GDP with GTP in the alpha subunit of G proteins. GTP binding modifies the conformation of G alpha proteins and activates them. The Regulator of G protein Signalling (RGS) proteins are GTPase Accelerating Proteins (GAPs) that can inhibit the G alpha subunit activity via their GAP activity. There are at least 25 different types of RGS proteins known. Several of these RGS proteins (1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21) can bind and stabilize the transition state for GTP hydrolysis of Guanine nucleotide binding protein G(i) subunit alpha (GNAI). Subsequently, this leads to GTP hydrolysis and inactivation of GNAI and terminating downstream signalling (Neubig & Siderovski DP 2002, Kach et al. 2012). The primary function of GNAI is the inhibition of adenylate cyclase.

Preceded by: [The Ligand:GPCR:Gi complex dissociates](#)

Followed by: [G alpha \(i\)-i1/i2/i3 in G \(i\):RGS complex is inactivated](#)

Literature references

Soundararajan, M., Willard, FS., Kimple, AJ., Turnbull, AP., Ball, LJ., Schoch, GA. et al. (2008). Structural diversity in the RGS domain and its interaction with heterotrimeric G protein alpha-subunits. *Proc. Natl. Acad. Sci. U.S.A.*, 105, 6457-62. ↗

Neubig, RR., Siderovski, DP. (2002). Regulators of G-protein signalling as new central nervous system drug targets. *Nat Rev Drug Discov*, 1, 187-97. ↗

Editions

2017-07-27	Authored, Edited	Varusai, TM.
2018-09-03	Reviewed	Siderovski, D.

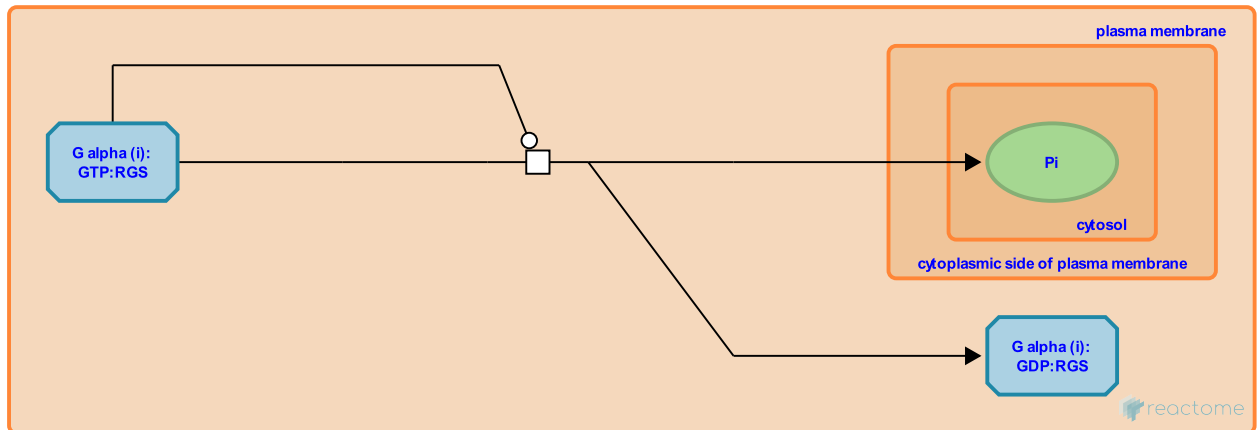
G alpha (i)-i1/i2/i3 in G (i):RGS complex is inactivated ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8982020

Type: transition

Compartments: plasma membrane



G Protein Coupled Receptors (GPCR) sense extracellular signals and activate different Guanine nucleotide binding proteins (G proteins). Upon activation, GPCRs can replace the GDP with GTP in the alpha subunit of G proteins. GTP binding modifies the conformation of G alpha proteins and activates them. The Regulator of G protein Signalling (RGS) are GTPase Accelerating Proteins (GAPs) that can directly inhibit the G alpha subunit activity. There are at least 25 different types of RGS proteins known. Several of these RGS proteins (1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21) can bind and stabilize the transition state of Guanine nucleotide binding protein G(i) subunit alpha (GNAI). Following this, the RGS domain of the proteins exert GAP activity on GNAI and allosterically modulate residues within G-alpha subunit to accelerate the intrinsic GTPase activity that hydrolyses GTP to GDP. This inactivates GNAI and terminates downstream signalling (Neubig & Siderovski 2002, Kach et al. 2012).

Preceded by: [Active G alpha \(i\)-i1/i2/i3 binds RGS proteins](#)

Followed by: [G alpha \(i\):RGS dissociates to inactive G alpha \(i\)-i1/i2/i3](#)

Literature references

Soundararajan, M., Willard, FS., Kimple, AJ., Turnbull, AP., Ball, LJ., Schoch, GA. et al. (2008). Structural diversity in the RGS domain and its interaction with heterotrimeric G protein alpha-subunits. *Proc. Natl. Acad. Sci. U.S.A.*, 105, 6457-62. ↗

Neubig, RR., Siderovski, DP. (2002). Regulators of G-protein signalling as new central nervous system drug targets. *Nat Rev Drug Discov*, 1, 187-97. ↗

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2018-09-03	Reviewed	Siderovski, D.

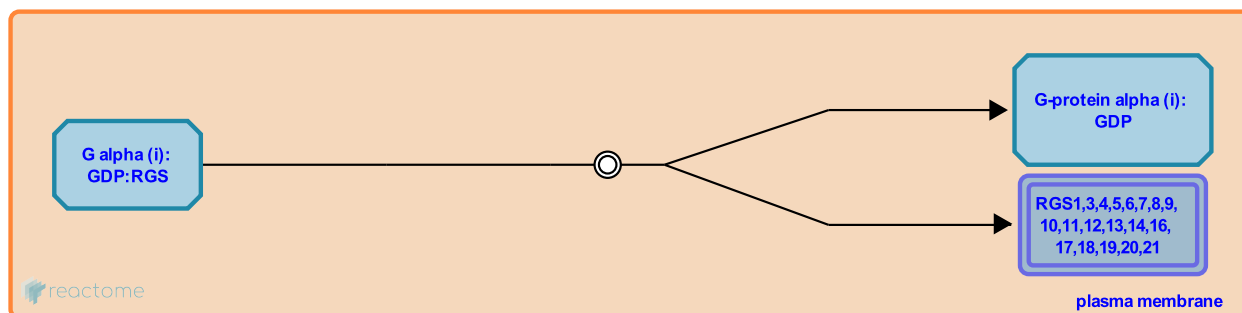
G alpha (i):RGS dissociates to inactive G alpha (i)–i1/i2/i3 ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8982012

Type: dissociation

Compartments: plasma membrane, cytosol



G Protein Coupled Receptors (GPCR) sense extracellular signals and activate different Guanine nucleotide binding proteins (G proteins). Upon activation, GPCRs can replace the GDP with GTP in the alpha subunit of G proteins. GTP binding modifies the conformation of G alpha proteins and activates them. The Regulator of G protein Signalling (RGS) are GTPase Accelerating Proteins (GAPs) that can directly inhibit the G alpha subunit activity. There are at least 25 different types of RGS proteins known. Several of these RGS proteins (1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21) can bind and stabilize the transition state of Guanine nucleotide binding protein G(i) subunit alpha (GNAI). Subsequently, RGS proteins in the complex facilitate the hydrolyses of GNAI:GTP to GNAI:GDP. Following this, the complex dissociates releasing inactive GNAI (Neubig & Siderovski 2002, Kach et al. 2012).

Preceded by: [G alpha \(i\)–i1/i2/i3 in G \(i\):RGS complex is inactivated](#)

Literature references

Soundararajan, M., Willard, FS., Kimple, AJ., Turnbull, AP., Ball, LJ., Schoch, GA. et al. (2008). Structural diversity in the RGS domain and its interaction with heterotrimeric G protein alpha-subunits. *Proc. Natl. Acad. Sci. U.S.A.*, 105, 6457-62. ↗

Neubig, RR., Siderovski, DP. (2002). Regulators of G-protein signalling as new central nervous system drug targets. *Nat Rev Drug Discov*, 1, 187-97. ↗

Editions

2017-07-27	Authored, Edited	Varusai, TM.
2018-09-03	Reviewed	Siderovski, D.

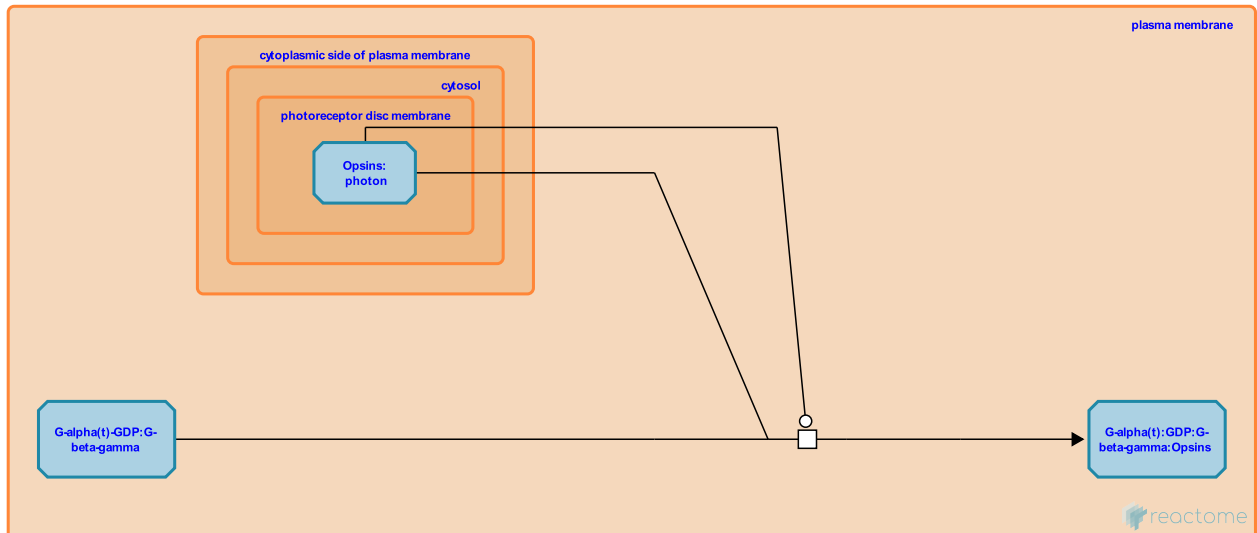
Opsins binds G alpha-t ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8982637

Type: transition

Compartments: plasma membrane



Transducin (Gt) is a heterotrimeric G protein encoded by GNAT genes and is naturally expressed in vertebrate retina rods and cones. There are two types, alpha-1 chain (expressed in rods by GNAT1) (Lerea CL et al, 1989) and alpha-2 chain (expressed in cones by GNAT2) (Morris TA and Fong SL, 1993). Photon activated-opsins can bind to G(t) alpha subunits and stimulate them. Activation of the G(t) alpha subunit results in the "vertebrate phototransduction cascade" (Chen CK, 2005). Cyclic GMP Phosphodiesterase is activated which lowers cGMP levels (an intracellular second messenger molecule). Lower cGMP levels can then lead to the closure of cGMP-regulated Na⁺ and Ca²⁺ ion channels and a hyperpolarized membrane potential.

Followed by: [Opsins act as GEFs for G alpha-t](#)

Literature references

Lerea, CL., Bunt-Milam, AH., Hurley, JB. (1989). Alpha transducin is present in blue-, green-, and red-sensitive cone photoreceptors in the human retina. *Neuron*, 3, 367-76. ↗

Morris, TA., Fong, SL. (1993). Characterization of the gene encoding human cone transducin alpha-subunit (GNAT2). *Genomics*, 17, 442-8. ↗

Chen, CK. (2005). The vertebrate phototransduction cascade: amplification and termination mechanisms. *Rev Physiol Biochem Pharmacol*, 154, 101-21. ↗

Editions

2009-05-14	Authored	Jassal, B.
2009-05-29	Reviewed	D'Eustachio, P.
2017-05-11	Edited	Jupe, S.
2017-07-10	Revised	Varusai, TM.

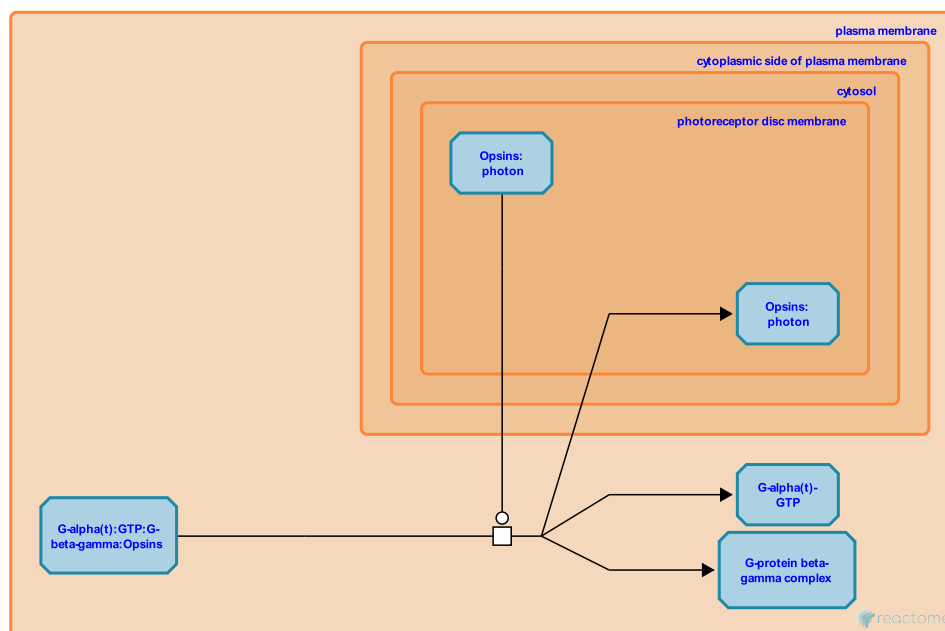
G-alpha(t):G-beta-gamma:Opsins complex dissociates to active G-alpha(t) ↗

Location: [G alpha \(i\) signalling events](#)

Stable identifier: R-HSA-8982640

Type: transition

Compartments: plasma membrane



Transducin (Gt) is a heterotrimeric G protein encoded by GNAT genes and is naturally expressed in vertebrate retina rods and cones. There are two types, alpha-1 chain (expressed in rods by GNAT1) (Lerea CL et al, 1989) and alpha-2 chain (expressed in cones by GNAT2) (Morris TA and Fong SL, 1993). Stimulated opsins can bind to and act as GEFs for G (t) alpha subunits thereby replacing the GDP with GTP. Subsequently, activated G (t) alpha proteins dissociate from the complex. Activation of the G (t) alpha subunit results in the "vertebrate phototransduction cascade" (Chen CK, 2005). Cyclic GMP Phosphodiesterase is activated which lowers cGMP levels (an intracellular second messenger molecule). Lower cGMP levels can then lead to the closure of cGMP-regulated Na⁺ and Ca²⁺ ion channels and a hyperpolarized membrane potential.

Preceded by: [Opsins act as GEFs for G alpha-t](#)

Literature references

- Lerea, CL., Bunt-Milam, AH., Hurley, JB. (1989). Alpha transducin is present in blue-, green-, and red-sensitive cone photoreceptors in the human retina. *Neuron*, 3, 367-76. ↗
- Morris, TA., Fong, SL. (1993). Characterization of the gene encoding human cone transducin alpha-subunit (GNAT2). *Genomics*, 17, 442-8. ↗
- Chen, CK. (2005). The vertebrate phototransduction cascade: amplification and termination mechanisms. *Rev Physiol Biochem Pharmacol*, 154, 101-21. ↗

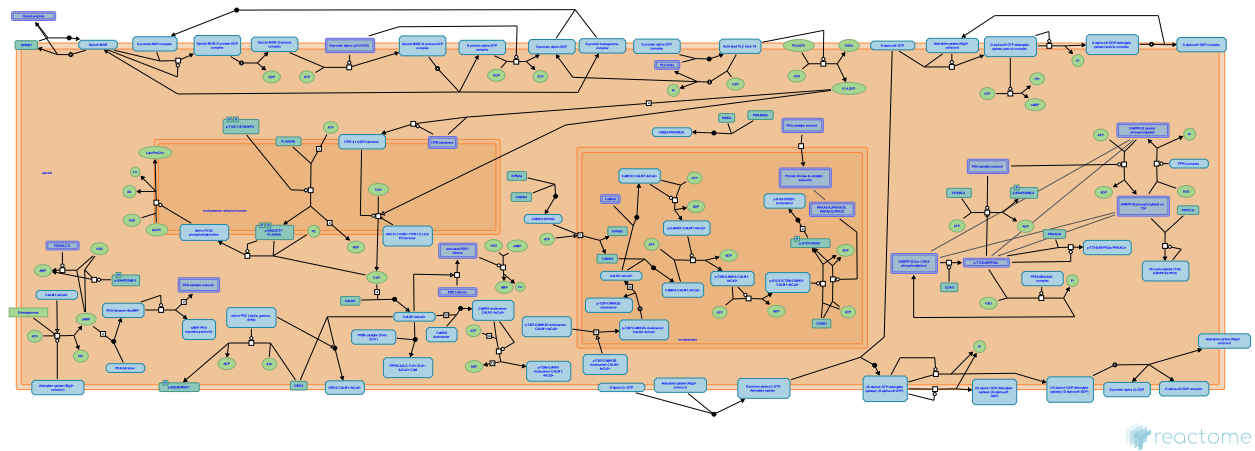
Editions

2009-05-14	Authored	Jassal, B.
2009-05-29	Reviewed	D'Eustachio, P.
2017-05-11	Edited	Jupe, S.
2017-07-10	Revised	Varusai, TM.

Opioid Signalling ↗

Location: G alpha (i) signalling events

Stable identifier: R-HSA-111885



Opioids are chemical substances similar to opiates, the active substances found in opium (morphine, codeine etc.). Opioid action is mediated by the receptors for endogenous opioids; peptides such as the enkephalins, the endorphins or the dynorphins. Opioids possess powerful analgesic and sedative effects, and are widely used as pain-killers. Their main side-effect is the rapid establishment of a strong addiction. Opioids receptors are G-protein coupled receptors (GPCR). There are four classes of receptors: mu (MOR), kappa (KOR) and delta (DOR), and the nociceptin receptor (NOP).

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Martin-Kleiner, I., Balog, T., Gabrilovac, J. (2006). Signal transduction induced by opioids in immune cells: a review. *Neuroimmunomodulation*, 13, 1-7. ↗

Standifer, KM., Pasternak, GW. (1997). G proteins and opioid receptor-mediated signalling. *Cell Signal*, 9, 237-48. ↗

Hamm, HE. (1998). The many faces of G protein signaling. *J Biol Chem*, 273, 669-72. ↗

Editions

2004-03-31	Authored	Jassal, B., Le Novere, N.
2008-11-06	Reviewed	Castagnoli, L.
2025-11-12	Edited	Schmidt, EE.

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